

OCR Further Mathematics
Pure Core - Proof
Question Paper
AS and A Level
Further Mathematics A - H235, H245



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The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.

1(a)

The matrix A is given by $A = \begin{pmatrix} 2 & 3 \\ 0 & 2 \end{pmatrix}$.

By considering A , A^2 , A^3 and A^4 make a conjecture about the form of the matrix A^n in terms of n for $n \geq 1$. [2]

(b) Use induction to prove the conjecture made in above part. [4]

2 Prove by induction that the sum of the cubes of three consecutive positive integers is divisible by 9. [5]



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3 Prove by induction that, for all positive integers n , $7^n + 3^{n-1}$ is a multiple of 4.

[5]



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4

Three matrices, A , B and C , are given by $A = \begin{pmatrix} 1 & 2 \\ a & -1 \end{pmatrix}$, $B = \begin{pmatrix} 2 & -1 \\ 4 & 1 \end{pmatrix}$ and $C = \begin{pmatrix} 5 & 0 \\ -2 & 2 \end{pmatrix}$ where a is a constant.

- (a) Using A , B and C in that order demonstrate explicitly the associativity property of matrix multiplication.[4]



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- (b) Use A and C to disprove by counterexample the proposition 'Matrix multiplication is commutative'. [2]

For a certain value of a , $\mathbf{A} \begin{pmatrix} x \\ y \end{pmatrix} = 3 \begin{pmatrix} x \\ y \end{pmatrix}$.

(c) Find

- y in terms of x ,
- the value of a .

[3]

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- 5 A sequence is defined by $a_1 = 6$ and $a_{n+1} = 5a_n$ for $n \geq 1$.

Prove by induction that for all integers $n \geq 1$, $a_n = \frac{11 \times 5^{n-1} + 1}{2}$. [5]

- 6 Prove by mathematical induction that, for all integers $n \geq 1$, $n^5 - n$ is divisible by 5. [4]



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- 7 Prove by induction that, for all positive integers n ,

$$\sum_{r=1}^n \frac{5-4r}{5^r} = \frac{n}{5^n}.$$

[5]

8 The sequence $u_1, u_2, u_3, \dots$ is defined by $u_n = 5^n + 2^{n-1}$.

(i) Find u_1, u_2 and u_3 .

[2]

(ii) Hence suggest a positive integer, other than 1, which divides exactly into every term of the sequence.

[1]

(iii) By considering $u_{n+1} + u_n$, prove by induction that your suggestion in part (ii) is correct.

[5]

9

The sequence $u_1, u_2, u_3, \dots$ is defined by $u_1 = 2$ and $u_{n+1} = \frac{u_n}{1 + u_n}$ for $n \geq 1$.

(i) Find u_2 and u_3 , and show that $u_4 = \frac{2}{7}$.

[3]

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(ii) Hence suggest an expression for u_n .

[2]

(iii) Use induction to prove that your answer to part (ii) is correct.

[5]

10 The sequence $u_1, u_2, u_3, \dots$ is defined by

$$u_1 = 5 \text{ and } u_{n+1} = 3u_n + 2 \text{ for } n \geq 1.$$

Prove by induction that $u_n = 2 \times 3^n - 1$.

[4]

11

$$\sum_{r=1}^n r(3r+1) = n(n+1)^2$$

Prove by induction that, for $n \geq 1$,

[5]

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12

The matrix M is given by $M = \begin{pmatrix} 2 & 2 \\ 0 & 1 \end{pmatrix}$. Prove by induction that, for $n \geq 1$,

$$M^n = \begin{pmatrix} 2^n & 2^{n+1} - 2 \\ 0 & 1 \end{pmatrix}.$$

[6]

END OF QUESTION PAPER



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